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EMS502U
Thermodynamic Cycles

Analysis of Geothermal Binary Power Cycles Utilising both SRC and ORC

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1 Introduction

Steam Rankine Cycles (SRCs) have been widely used in power generation over history, ranging from coal powered to more modern approaches utilising nuclear heat generation. However, since the 1950s, when Organic Rankine Cycles (ORCs) were developed, a focus on implementing these as a form of heat recovery was established after determining their applicability in utilising low to medium temperature heat sources alongside high molecular mass organic fluids as the working fluid to generate electricity. This paper focuses on the analysis and comparison of SRCs against ORCs in geothermal binary powerplants, with an ultimate goal to conclude which is superior when working with low grade heat.

Geothermal energy is a renewable form of energy stored as heat within the Earth's interior, and both the natural steam and high temperature water form of it have been exploited for decades in many industrial processes (Barbier, 2002). However, unlike fossil fuels, geothermal sources provide a continuous supply of energy, reducing the dependency on non-renewable resources and promoting sustainability, additionally it is estimated that the energy stored at depth of 3 km from the Earth's surface is as high as $43,000,000 \times 10^{18}$ J (DiPippo, 2016). In 2000, the geothermal electrical installed capacity in the world was 7974 MWe (as seen in Figure 1) and the electrical energy generated was 49.3 billion kWh/year, representing 0.3 % of the world's total electrical energy which was 15,342 billion kWh. An increase of about 2GW was achieved in the five year term 2005–2010, with an estimate that the installed capacity of geothermal powerplants could make up 8.3% of total world electricity production by 2050 (Bertani, 2012). In recent times, ORC has raised great interest as it makes it possible to produce electricity from cooler geothermal sources, typically within the 80–150°C temperature range (Herath et al., 2020), often available from below 1000m (Cammarata, Cammarata and Petrone, 2014). Furthermore, the condenser temperature and pressure in geothermal binary powerplants can range from 10-40°C (Ghasemi et al., 2013), and standard atmospheric pressure of 1 bar, respectively. The standard first and second law efficiencies for geothermal binary plants were discovered to be approximately 5-15% (DiPippo, 2004) and 20-54% respectively (Franco and Villani, 2009), with the power output estimates around 2MW (Paweł Ziółkowski et al., 2021). The upcoming discussion will detail a critical analysis between SRCs and ORCs, as well as potential working fluid options for use within geothermal binary powerplants.

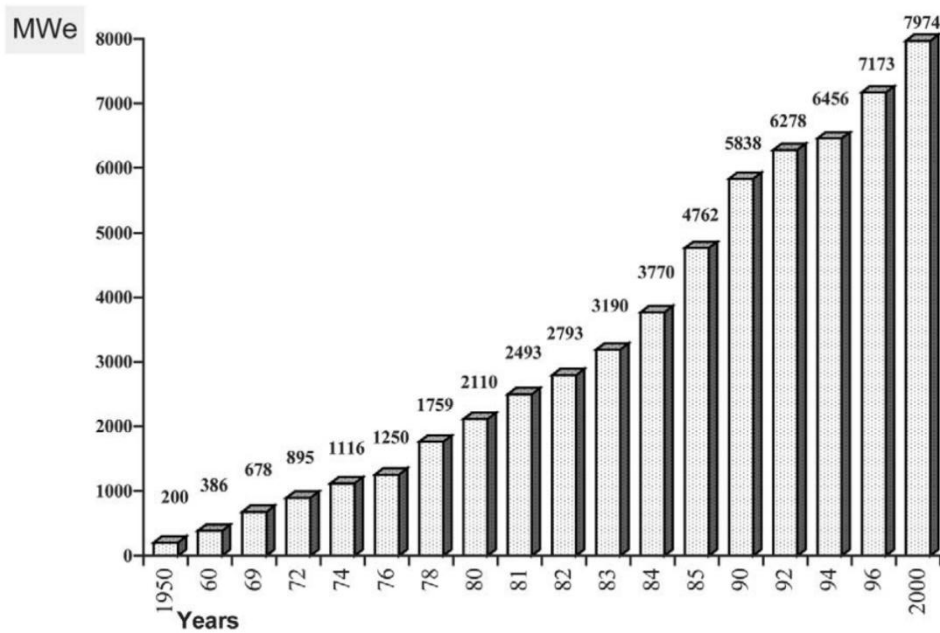


Figure 1: Illustration showing the increasing prevalence of geothermal power generation around the world during the period ranging from 1950-2000 (Barbier, 2002)

2 Analysis of SRCs and ORCs

2.1 Common types of working fluids

The types of working fluids used in powerplants can be classified into 3 groups: wet, dry, and isentropic. They can be distinguished via their slopes on a (Temperature-entropy) T-s diagram, where the slope of the saturation vapour curve is $\xi = \left(\frac{ds}{dT}\right)$, where for wet fluids $\xi < 0$, dry fluids $\xi > 0$, and isentropic fluids $\xi \approx 0$ (Liu, Chien and Wang, 2004). A key difference in the use of water and common organic fluids is the fact that water is a wet fluid, whereas most organic fluids used within power cycles are either dry or isentropic, this presents potential difficulties for SRC operation, since the water saturation curve has a negative gradient (as depicted in Figure 2(a)) requiring the SRC to preferably always include a superheating stage. This is due to the limiting temperature of 85-150°C and low evaporator pressures around 5–10 bar in a geothermal binary powerplant, making it near impossible for the water, which has a relatively high boiling point of 100°C, to be completely converted to a saturated gaseous state, due to its critical temperature being 374.15°C (Dunham and Iverson, 2014). This would be necessary to limit the amount of wet fluid droplets present within the turbine during the expansion stage, minimising damage to the blades, consequently, the costs for regular maintenance significantly increases, alongside the costs for the inclusion of advanced superheating equipment (Tchanche et al., 2011). Contrarily, dry and isentropic fluids have either positive gradients or no gradient in their saturation curves (Figures 2(b) and 2(c)), eliminating the need for expensive superheaters and complex insulation needed within the cycle as they do not produce wet droplets, due to this, the turbine blades would require less maintenance. However for dry fluids, a different issue arises with the fact that the fluid exits the turbine at a superheated state, hence why a recuperator is often used to cool the vapor before reaching the condenser, and to preheat the fluid before the evaporator, which can slightly improve efficiency, and offset the cost for its inclusion (Groniewsky, Györke and Imre, 2017). The added maintenance costs over the lifetime of an SRC plant, would make it less financially viable for utilising low grade heat, justifying the use of ORCs as a better alternative in such cases with lower operating temperatures and pressures.

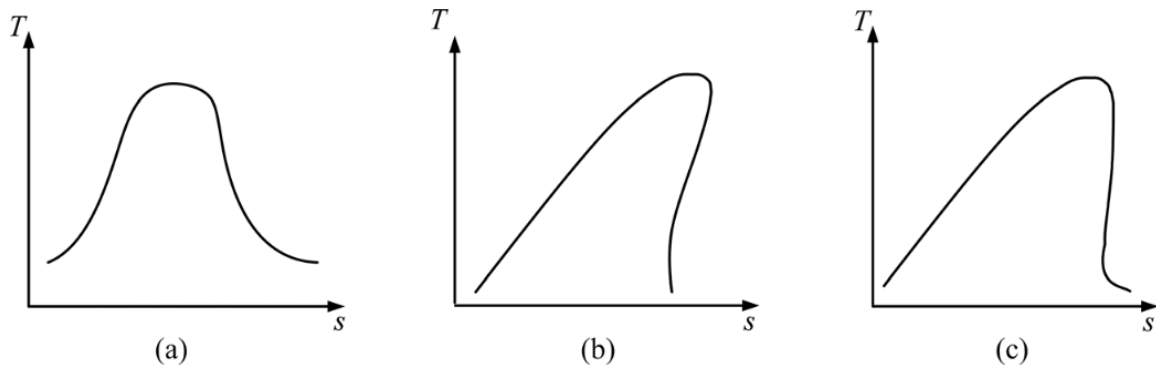


Figure 2: Temperature-entropy diagram for different working fluid classes - (a) wet, (b) dry, and (c) isentropic

Furthermore, Amicabile, Lee and Kum (2015) detailed eleven basic guidelines for screening suitable working fluids, adapted by Shalby, Abdullah Marachli and Salah (2025) highlighted in Table 1 below.

Table 1: Working fluids screening criteria (Shalby, Abdullah Marachli and Salah, 2025)

Criteria	Description
Thermodynamic performance	Dry or isentropic fluids preferred to wet, to avoid turbine damage. High efficiency affected by properties such as specific heat and density.
Environmental impact	Zero Ozone Depletion Potential, low Global Warming Potential, and short atmospheric lifespans to comply with environmental regulations.
Safety	Low flammability, non-toxic, non-explosive, and easy to handle.
Operating conditions	Should exhibit acceptable evaporating and condensing pressures (above 0.1MPa for condensers and above 3MPa for evaporators), low viscosity for reduced frictional losses, and good thermal conductivity for efficient heat transfer. Critical temperature slightly higher than the expander's inlet temperature.
Chemical stability	Thermally stable up to 350°C, to prevent deterioration or degradation.
Availability and cost	Easily accessible, cost-effective, low specific volumes for efficient utilisation and commercialisation.
Latent heat and cost	High latent heat and density to absorb more energy from the evaporator when dealing with low temperature heat sources.
Lubricant compatibility	Must be soluble with lubricant oils used within the system.
Freezing point	Melting point should be low enough to prevent freezing during operation in cold environments.
Molecular weight	Larger molecular weight fluids reduce flow rate and plant size, leading to improved turbine efficiency.
Boiling temperature	Appropriate boiling point under operating pressures, enabling thermal energy recovery from low-grade heat sources.

2.2 Geographic limitations and availability

An additional factor to consider would be the location constraints for either cycle, as water is a necessity for SRCs, therefore the plants need to be built around large water reservoirs, which does not pose much of an issue due to its high availability and low cost. Conversely, ORCs have the benefit of being more versatile, since they can be utilised in a range of different locations and applications, such as solar, biomass, etc. However, a glaring issue with the use of ORC is that it must be built a safe distance from highly populated areas due to the potential risk and hazards posed by the properties of organic fluids (Raab, Klein and Opferkuch, 2021), since they can be toxic, potentially flammable, or environmentally impactful (a summary comparison can be seen in Table 2). Furthermore, organic fluids pose a difficulty in sourcing, as there are many regulations in place that increasingly limit the usage of particular fluids with a high GWP and ODP, and traditional refrigerants are also extremely expensive, unless mass produced, making them less ideal than water in that regard (Chen, Yogi and Stefanakos, 2010). By contrast, water is stable, non-toxic, non-polluting, and readily available, plus the high thermal conductivity of water results in an increased efficiency (Chowdhury and Monjurul, 2023).

These differences in availability, specialised equipment required, and the location constraints, produce a significant difference in cost between the SRC and the ORC. Since geothermal plants preferably require a much simpler design and much less space they are ideal for power generation utilising the low grade heat sources, and SRCs would be significantly less efficient in such a scenario where low power generation is required, as such demanding conditions are required, there would be

diminishing returns. This aspect will be further discussed in the following section, where a detailed comparison of SRCs and ORCs operating conditions is conducted.

Table 2: Brief comparison of SRC and ORC properties (Tchanche et al., 2011)

	Steam Rankine Cycle	Organic Rankine Cycle
Fluid	Water	Organic compound
Critical pressure	High	Low
Critical temperature	High	Low
Boiling point	High	Low
Condensing pressure	Low	Acceptable
Specific heat	High	Low
Viscosity	Low	Relatively high
Flammability	No	Yes, and depends on fluid
Toxicity	No	Yes
Environmental impact	No	High and depends on fluid
Availability	Available	Supply problem
Cost	Cheap	Expensive

2.3 Operating conditions and functional differences

Functionally, SRCs and ORCs are vastly different, this is mainly due to the difference in working fluid used, and the conditions they best operate in. Generally, SRCs have to be run at high temperatures of around 500°C or higher, and excessive evaporator pressures upwards of 17MPa (Markides and Wang, 2023) for the fluid to reach and maintain the critical gaseous state to avoid condensation during expansion within the turbine, which is why they require a superheating stage to satisfy the temperature requirement (Raab, Klein and Opferkuch, 2021). This was further confirmed by Hung, Shai and Wang (1997), who found that a SRC does not allow for efficient recovery of low grade heat below 370°C. These excessive operating conditions and specialised equipment required would cause the plant to increase in size drastically, as well as increase component costs due to the sophisticated safety procedures needed (Baru et al., 2016), hence why, they are most often run using nuclear and various fossil fuels as the heat source, which can provide and maintain these extreme temperature requirements, consequently making these powerplants significantly larger and expensive to build and maintain. Furthermore, SRCs require the removal of air within the condenser, as its presence can cause the steam saturation temperature to lower, or the effect of air blanketing can cause the heat transfer in the system to drastically reduce, therefore, to implement such a feature would result in a bulkier system design and increased costs (Dyke, 2005). Alternatively, the binary geothermal ORC (Figure 3), can effectively operate at significantly reduced evaporator conditions of up to and around 150°C and 5 bar, and condenser pressures of 1 atm since most organic fluids have a relatively lower boiling point compared to water, allowing them to easily reach the saturated gaseous state. Furthermore, the smaller temperature difference between the evaporator and condenser is instrumental for simpler single stage turbines to be feasible, since the pressure ratio will also be smaller (Tchanche et al., 2011).

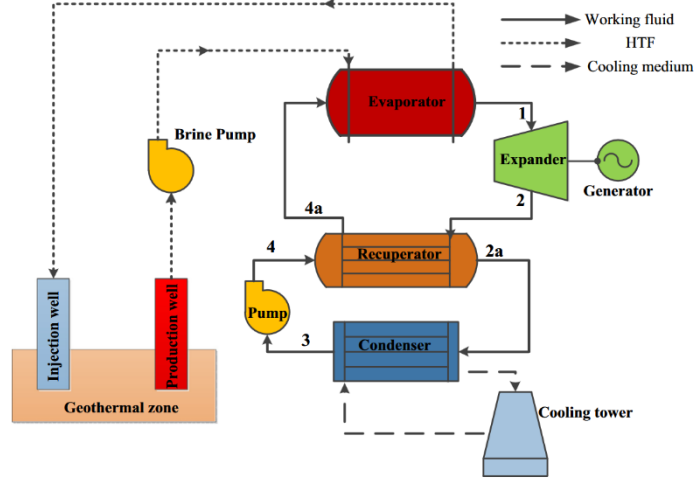


Figure 3: Schematic of a geothermal binary power cycle (ORC)
(Rahbar et al., 2017)

Water is used as the working fluid in SRCs due to it inherently possessing a high latent heat of vaporisation and specific heat, allowing it to carry a great deal of thermal energy in its gaseous form, improving efficiencies at higher temperatures, additionally, its low molecular mass leads to a lower mass flow rate, which is paramount in reducing the pump power required to pump the fluid through the system, whereas with organic fluids, the higher mass flow rate leads to a greater pump power required. Furthermore, the mass flow rate also affects the cost of the plant, since higher mass flow rates mean that more sophisticated materials would be required to support the system, drastically increasing the initial capital investment cost of the plant (Baru et al., 2016). The pump power can be characterised by Equation 1 (Brian and Brian, 2014) as follows,

$$\text{Pump power, } W = \frac{\dot{m} \cdot \Delta P}{\rho \cdot \eta_{\text{pump}}} \quad (1)$$

Where $\dot{m}$ is the mass flow rate, ΔP is the pressure difference, ρ is the density of the fluid, and η_{pump} is the efficiency of the pump.

Organic working fluids are preferable to water in order to effectively convert as much of the low grade heat into salvageable power as possible, which was discussed in depth by Chen, Yogi and Stefanakos (2010) and further verified by Bao and Zhao (2013), where they detailed how working fluids with a high density, low liquid specific heat and a high latent heat are expected to provide a greater work output. This was described through an equation by Chen, Yogi and Stefanakos (2010), showing that fluids with higher latent heat give higher work outputs, seen in Equation 2.

$$\Delta h_{\text{isen}} = C_p T'_{\text{in}} \left[1 - e^{L(1/T_1 - 1/T_2)/C_p} \right] \quad (2)$$

where T_1 and T_2 are the saturation temperatures of two points on the 2-phase line and $T_1 > T_2$, T'_{in} is the turbine inlet temperature.

The relatively higher molecular weight of organic fluids grants additional benefits to the system as well, since molecular weight is inversely proportional to the expansion work by turbines, it allows for lesser turbine rotations and as a result, lower vapour pressures, which can reduce the complexity and safety concerns of the system during operation (Rahbar et al., 2017; Bao and Zhao, 2013). As a consequence of this, less energy is required to be inputted into the system, allowing for an overall greater net output, improving efficiencies compared to the SRC. An additional benefit is that erosion of the turbine blades is minimised and smaller more compact plants can be developed. Furthermore, the specific volume of organic fluids is smaller than that of steam, hence the size parameter of SRC turbines are up to 2x larger than ORC turbines (Shu et al., 2014), significantly reducing the capital costs of building an ORC geothermal plant compared to the alternative. In the next section, the most suitable and effective ORC working fluids will be critically analysed to determine which is most optimal for use within the geothermal binary powerplant.

2.4 Working fluids used

As discussed in Section 2.1, the most optimal types of fluids for use within an ORC binary geothermal system would be either dry or isentropic, as they provide fewer downsides compared to wet fluids. From comparing 24 different working fluids, Wang et al. (2020) determined that those with a suitable critical temperature (T_{cri}) within a specific range of the heat source temperature, lower dryness (ξ), and lower critical pressure (P_{cri}) can achieve better overall power outputs. To add to this, Basaran and Leyla Ozgener (2013) compared various dry and wet type fluids within a binary geothermal powerplant model with a more general perspective of analysis, from which dry fluids were found to exhibit better efficiencies compared to wet fluids. Specifically, they found R600 (Butane) to produce the greatest first and second law efficiencies of 13.73% and 65.89% respectively, when the evaporator temperature and pressure were 90°C and 1MPa respectively, and the condenser at 10°C. Additionally, Sakhrich et al. (2016) investigated 16 working fluids, placing an emphasis on work and efficiency being the key output parameters; through this it was found that the fluids all exhibited positive results within a given temperature range, however, R600 was ultimately found to be the most suitable option for the application, corroborating Basaran and Leyla Ozgener (2013) by obtaining a first law efficiency of 14% and a work output of 50kJ/kg for a low temperature heat source. Chen et al. (2019) identified that working fluids had maximum exergy efficiency (η_{ex}) when the heat source temperature approached the critical temperature of the fluids, and through comparing 18 working fluids it was found that under a condenser and evaporator temperature of 40°C and 130°C respectively, R236ea exhibited optimal properties, however when the evaporator temperature was 150°C, R245fa exhibited optimal properties. This was further verified by Hu et al. (2022), where 32 working fluids were considered and it was found that R245fa was most optimal for when the system aims to maximise the net power produced per unit mass of geothermal water. Additionally, Wang, Liu and Zhang (2014) analysed the performance of 3 different working fluids at different temperatures and conditions, finding R245fa performed the best when optimising for thermodynamic and economic performance within an ORC system combined with orthogonal design. Zhang et al. (2011), investigated 16 working fluids and determined that R123 in a subcritical ORC system yielded the highest thermal and exergy efficiencies of 11.1% and 54.1% respectively. This was further substantiated by Liu et al. (2017), where it was found that R123 was optimal for thermal efficiency and capital cost performance when the heat source temperature increased from 80-95°C, however R134a was found exhibit better work output and exergy efficiency for conditions over 90°C. Madhawa et al. (2007) considered 4 working fluids and developed an objective function that describes the ratio of total heat exchanger area to net power output (Equation 3), which was optimised using the steepest descent method to evaluate the working fluid that allowed for minimum cost and maximum efficiency. It was evaluated that the fluids HCFC 123 (R123) and n-Pentane had better performances.

$$\text{Objective function, } \gamma = \frac{A_T}{W_N} \quad (3)$$

Where A_T is the total heat transfer area, and W_N is the total net power.

By contrast, Guo et al. (2011) conducted research on 13 working fluids in a transcritical cycle and found that from a thermodynamic and techno-economic point of view, R125 presented the best performance. It showed 10% higher net power output and 22% reduction of total heat transfer areas compared with baselines given geothermal source temperature of 90°C. With the geothermal source temperature above 100°C, and using pinch point analysis on a transcritical Rankine power cycle, they found R32 and R143a also show better performances. Mago et al. (2007) conducted research on 7 organic fluids to determine the effect of boiling point temperature on the performance of ORCs, where it was verified that organic fluids with a higher boiling point exhibited better efficiencies, correlating to the fact that organic fluids with higher molecular masses also provide better performances since molecular weight and boiling point have a directly proportional relationship. Comparing these fluids

within a low heat ORC cycle they found that R123, R245ca, and R245fa showed the best efficiencies for temperatures between 380 and 430 K; and isobutane for temperatures <380 K.

Critically analysing these selections of fluids, and considering all the suitability factors for safe and effective use within a powerplant, it can be determined that R600 (or Butane) is the optimal choice for effectively utilising the low grade heat from geothermal binary powerplants. This can be mainly attributed to Shalby et al. (2025) stating the fact that R245fa and R123, the two most promising candidates, have been prohibited due to their high GWP and ODP, furthermore, R134a is also in contention for banning in the near future due to similar reasons. Hence, the next most ideal option would be R600 since it is a pure fluid, as a result it has low environmental impact and its critical temperature being approximately in the range for geothermal sources allows for improved efficiency. It inherently being a dry fluid improves its effectiveness in an ORC, furthermore Basaran and Leyla Ozgener (2013) substantiated its performance, with further calculations performed in Section 3 to verify these claims. It can be seen compared to water in Table 3 below.

Table 3: Property comparisons between water and butane, adapted from (Shalby, Abdullah Marachli and Salah, 2025)

	Working fluids	
	Water (R718)	Butane (R600)
Molecular mass (kg/kmol)	18	58.12
T_{bp} (°C)	100	-0.5
T_{cr} (°C)	374	152
P_{cr} (MPa)	22.064	3.796
Atmospheric lifetime (years)	n.a.	0.018
ODP	0	0
GWP (100 yr)	<1	3
Expansion molecular complexity	wet	dry

3 Analysis of the chosen working fluid

3.1 Chosen operating conditions

Table 4: Assumed operating conditions for the following calculations (Basaran and Leyla Ozgener, 2013)

Operating conditions of the Geothermal Binary Power Plant	
Condenser Temperature, T_c (°C)	10.00
Condenser Pressure, Pa (MPa)	0.148
Evaporator Temperature, T_H (°C)	90.00
Condenser Pressure, Pa (MPa)	1.000
Mass flow rate, $\dot{m}$ (kg/s)	22.25
Turbine efficiency (%)	85
Pump efficiency (%)	75

3.2 Carnot cycle

The Carnot cycle (see Figure 4) is an ideal thermodynamic cycle that occurs completely within the 2 phase region and aims to maximise the efficiency of a given process. It describes that a process can be 100% efficient if a heat source with an infinitely high temperature and a heat sink at absolute zero temperature is utilised. This cycle consists of 2 isothermal and 2 isentropic stages, which are ideal and reversible, hence the efficiency only depends on the chosen operating conditions rather than depending on the choice of working fluid.

Carnot cycle efficiency, (η_{carnot}), is given by Equation 4:

$$\eta_{carnot} = 1 - \frac{T_C}{T_H} \quad (4)$$

$$\eta_{carnot} = 1 - \left(\frac{10 + 273}{90 + 273} \right) = 0.2204 \approx 22.0\%$$

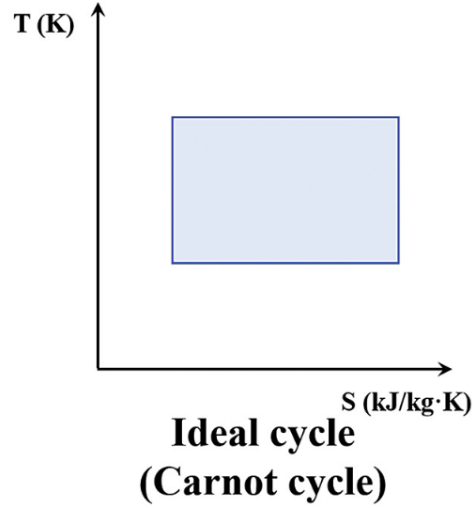


Figure 4: An ideal Carnot cycle visualised and adapted from (Xu et al., 2018)

3.2 Steam Rankine Cycle

The first law efficiency, which describes the quantity of energy that is conserved, and specific steam consumption are calculated for water (Figure 5) as the working fluid in the following part, the detailed calculations can be found in the Appendix (Section 6.1).

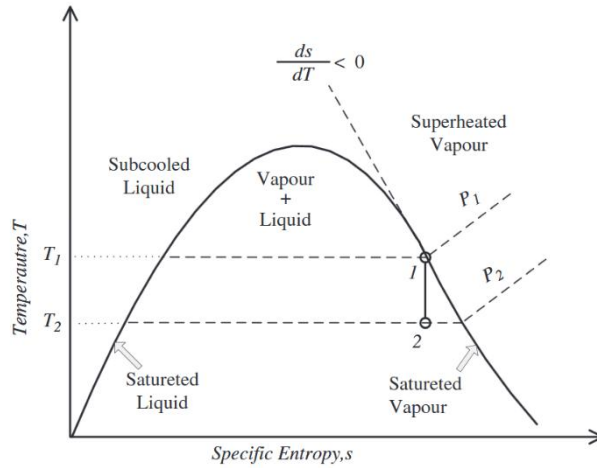


Figure 5: T-s diagram for water (wet fluid) adapted from (Bao and Zhao, 2013)

First law efficiency is given by the following equation (Equation 5):

$$\eta = \frac{-(\dot{W}_{pump} + \dot{W}_{turb})}{\dot{Q}_{in}} \quad (5)$$

$$\eta \text{ (for water)} = \frac{-(25.22 - 6279.90)}{51440.89} = 0.1216 \text{ (12.16\%)}$$

Specific Steam Consumption (SSC), is given by Equation 6:

$$SSC = \frac{3600 \times \dot{m}}{-(\dot{W}_{pump} + \dot{W}_{turb})} \quad (6)$$

$$SSC \text{ (for water)} = \frac{3600 \times 22.25}{-(25.22 - 6279.90)} = 12.81 \text{ kg/kWh}$$

The total power output from utilising water within the geothermal binary powerplant is given by the following (Equation 7):

$$\dot{W}_{net} = -(\dot{W}_{pump} - \dot{W}_{turb}) \quad (7)$$

Therefore,

$$\dot{W}_{net} = -(25.22 - 6279.90) = 6254.68 \text{ kW}$$

3.3 Organic Rankine Cycle (using Butane or R600)

The first law efficiency and specific steam consumption are calculated for butane (Figure 6) as the working fluid in the following part, the detailed calculations can be found in the Appendix. The values for this calculation were obtained from Basaran and Leyla Ozgener (2013) and verified by govinfo.gov (1982). The values for enthalpy through the pump and turbine stages already had the isentropic efficiencies applied.

Similarly to the previous calculation, efficiency, η , is given by:

$$\eta \text{ (for R600)} = \frac{-(42.94 - 1555.05)}{11010.64} = 0.1373 \text{ (13.73\%)}$$

Additionally, Specific Steam Consumption (SSC), is given by:

$$SSC \text{ (for R600)} = \frac{3600 \times 22.25}{-(42.94 - 11010.64)} = 52.97 \text{ kg/kWh}$$

Furthermore, the total power output of an ORC utilising R600 is:

$$\dot{W}_{net} = -(42.94 - 1555.05) = 1512.11 \text{ kW}$$

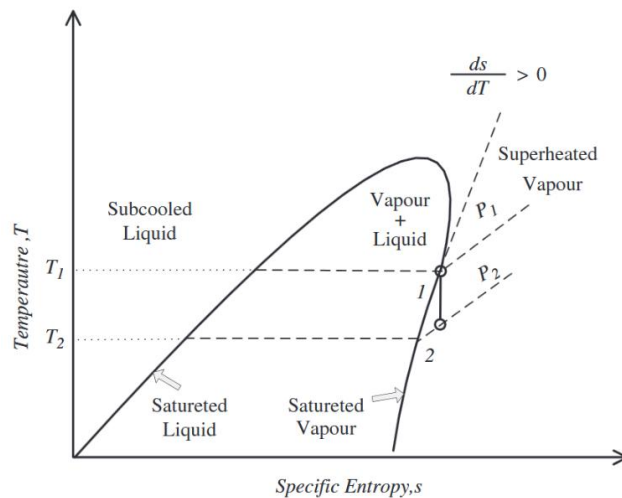


Figure 6: T-s diagram for Butane (or R600), a dry fluid, adapted from (Bao and Zhao, 2013)

4 Conclusions

As discussed previously, it has been determined that ORCs are superior to SRCs within a geothermal binary powerplant, this can be surmised from the some fundamental differences that exist between them. Some of the main factors include the fact that SRCs utilise water, which being a wet fluid possesses less favourable traits, due to the potential of wet droplets damaging the turbine blades and lowering the overall efficiency of the system. This shortcoming can be overcome by considering an organic fluid as the alternative, however they also come with various downsides, including availability constraints. This is proven by the extensive literature review, detailing and comparing SRC against ORC, showing that SRCs cost significantly more to build and maintain. Additionally, the calculations performed, comparing R600 and water show the ORC to have an overall better efficiency, 13.73% compared to 12.16%, when handling the specified conditions of the geothermal binary power plant, meaning that while SRCs produce a higher power output for the given conditions, if the same operation was upscaled, the higher efficiency of the ORC would outweigh the SRC. Furthermore, the higher SSC of the ORC can be explained due to the relatively higher density compared to water, hence the powerplant would not require an increased area to be built. Therefore, through the thorough analysis performed in Section 2 and Section 3, it can be deduced that ORCs exhibit outstanding performance when handling low grade heat sources, and are the better option for a functioning geothermal binary powerplant.

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6 Appendix

6.1 Extended SRC calculations

Utilising Table 5, the h_f and h_g values for the isobar at 1.48 bar can be determined, and that of 10 bar can be obtained from the table.

Then, through following standard calculation procedures, h_1 , h_{1a} , h_3 , and h_4 can be calculated as follows.

$$h_1 = h_f(\text{at } 1.48 \text{ bar}) = 465.2 \text{ kJ/kg}$$

$$h_{1a} = h_1 + v_f(P_{1a} - P_1) = 465.2 + 1 \times 10^{-3}(10 - 1.48) \times 10^2 = 466.05 \text{ kJ/kg}$$

$$h_3 = h_g(\text{at } 10 \text{ bar}) = 2778 \text{ kJ/kg}$$

$$h_4 = h_f(1.48 \text{ bar}) + x h_{fg}(1.48 \text{ bar}) = 465.2 + \left(\frac{6.586 - 1.4294}{7.2276 - 1.4294} \right) \times 2227.2 = 2445.95 \text{ kJ/kg}$$

Therefore, using the methods displayed by Wang, Liu and Zhang (2014):

$$\dot{W}_{turb} = \dot{m}(h_4 - h_3) \times 0.85 = -6279.90 \text{ kW}$$

$$\dot{W}_{pump} = \frac{\dot{m}(h_{1a} - h_1)}{0.75} = 25.22 \text{ kW}$$

$$\dot{Q}_{in} = \dot{m}(h_3 - h_{1a}) = 51440.89 \text{ kW}$$

Table 5: Properties table for saturated water and steam (Rogers and Mayhew, 1995)

Saturated Water and Steam

p [bar]	T_s [°C]	v_g [m ³ /kg]	u_f [kJ/kg]	u_g [kJ/kg]	h_f [kJ/kg]	h_{fg} [kJ/kg]	h_g [kJ/kg]	s_f [kJ/kg K]	s_{fg} [kJ/kg K]	s_g [kJ/kg K]
1.0	99.6	1.694	417	2506	417	2258	2675	1.303	6.056	7.359
1.1	102.3	1.549	429	2510	429	2251	2680	1.333	5.994	7.327
1.2	104.8	1.428	439	2512	439	2244	2683	1.361	5.937	7.298
1.3	107.1	1.325	449	2515	449	2238	2687	1.387	5.884	7.271
1.4	109.3	1.236	458	2517	458	2232	2690	1.411	5.835	7.246
1.5	111.4	1.159	467	2519	467	2226	2693	1.434	5.789	7.223
10	179.9	0.1944	762	2584	763	2015	2778	2.138	4.448	6.586

6.2 Extended ORC calculations

Through Table 6 it can be seen that the enthalpy values for the geothermal binary cycle utilising R600 have been precalculated by Basaran and Leyla Ozgener (2013), which were further verified by govinfo.gov (1982). Only paying attention to states 1, 2, 3, and 4, the operating conditions and corresponding enthalpy and entropy states. These calculations include the turbine and pump efficiencies of 0.85 and 0.75 respectively.

Table 6: calculated thermodynamic properties for R600 (Basaran and Leyla Ozgener, 2013)

State number	Working fluid	Phase	T (K)	P (MPa)	h (kJ kg ⁻¹)	s (kJ kg ⁻¹ K ⁻¹)	$\dot{m}$ (kg s ⁻¹)
0	Brine	Dead state	278.00	0.101	20.49	0.07398	–
0'	Cooling water	Dead state	278.00	0.101	20.49	0.07398	–
0''	R 600	Dead state	278.00	0.101	597.23	2.73010	–
1	R 600	Saturated liquid	283.00	0.148	223.04	1.08260	22.25
2	R 600	Liquid	283.53	1.000	224.97	1.08430	22.25
3	R 600	Superheated vapor	363.00	1.000	719.83	2.54170	22.25
4	R 600	Superheated vapor	311.95	0.148	649.94	2.58170	22.25
5	Brine	Liquid	369.00	0.500	401.98	1.25980	75.00
6	Brine	Liquid	334.00	0.500	255.14	0.84170	75.00
7	Cooling water	Liquid	280.00	0.500	29.29	0.10410	283.25
8	Cooling water	Liquid	288.00	0.500	62.83	0.22221	283.25

$$h_1 = 223.04 \text{ kJ/kg}$$

$$h_2 = 224.97 \text{ kJ/kg}$$

$$h_3 = 719.83 \text{ kJ/kg}$$

$$h_4 = 649.94 \text{ kJ/kg}$$

$$\dot{W}_{turb} = \dot{m}(h_4 - h_3) = -1555.05 \text{ kW}$$

$$\dot{W}_{pump} = \dot{m}(h_2 - h_1) = 42.94 \text{ kW}$$

$$\dot{Q}_{in} = \dot{m}(h_3 - h_2) = 11010.64 \text{ kW}$$